

Morse-Bott-Smale Theory

Lecture 0	Overview [24/7]
Topic	Morse Theory (MT)
Lecture 1	Morse lemmas, handle attachment, Morse inequalities [30/7]
Lecture 2	Morse inequalities; existence of Morse functions [6/8]
Lecture 3	Closed geodesics exist [13/8]
Lecture 4	Morse-Smale flows, stable and unstable submanifolds [20/8]
Lecture 5	Morse theory via currents I [27/8]
Lecture 6	Morse theory via currents II [3/9]
Topic	Morse-Bott Theory (MBT)
Lecture 7	Reeb's Theorem & exotic spheres [10/9]
Lecture 8	Generalized Reeb's Theorem [17/9]
Lecture 9	Definition & basic properties/examples [24/9]
Lecture 10	Cohomology of complex Grassmannians I [1/10]
Lecture 11	Cohomology of complex Grassmannians II [8/10]
Topic	Cohen-Jones-Segal Theorem (CJS)
Lecture 12	Flow category of Morse function [15/10]
Lecture 13	Simplicial sets & geometric realization [29/10]
Lecture 14	Bar construction of algebras & categories [5/11]
Lecture 15	Moduli space & classifying space of flow categories [12/11]
Lecture 16	Bar construction of infinity categories & algebras [19/11]
Topic	Further Topics (FT)
Lecture 17	G-equivariant Morse-Bott functions [26/11]
Lecture 18	Circle-equivariant (co)homology [3/12]
Lectures 19, 20	Morse-Bott theory on free loop spaces [10/12, 17/12]
Lectures 21, 22	Circle valued Morse functions via flat connections []
Lecture 23	Complex Morse theory []